

SEC P vs NP log 2026-05-12

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-12 07:00 UTC

SEC P vs NP — daily log 2026-05-12

15 cycles recorded

Free Cumulant Sum Bounded by Disjointness Communication Complexity

- **Verdict:** INCONCLUSIVE
- **Bridge:** Free Probability Theory $\times$ Communication Complexity of DISJOINTNESS
- **Recorded:** 2026-05-12 01:18 UTC
- **Entry ID:** ecd8f062c47d

Statement

For a $|X| \times |Y|$ matrix M defining a partial function f , let $\mu(M) = \sum_{k \geq 1} |c_k(M)|$ where $c_k(M)$ are the free cumulants of M 's entries. Then $\mu(M) \leq O(\sqrt{R(f)})$ where $R(f)$ is the randomized communication complexity of f . For DISJ_n , $\mu(\text{DISJ}_n) = \Omega(n)$ holds.

Rationale

Free cumulants capture non-commutative independence structures in matrix entries, which may correlate with the information-theoretic cost of coordinating disjointness checks. The sum's growth rate could reflect the minimal communication required to resolve entangled dependencies.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_38ecf086.py", line 75, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_38ecf086.py", line 54, in run_trial
    cumulants = free_cumulants(M)
                ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_38ecf086.py", line 39, in free_cumulant
    R[i][j] -= sum(R[x][y] * R[i - x][j - y] for x in range(i) for y in range(j)) / (i * j)
               ^~~~~~
```

ZeroDivisionError: division by zero

Judge reasoning

Test crashed due to division by zero, preventing evaluation of conjecture | next: Fix implementation to handle zero division and retest with validated inputs

Plethysm Coefficient Gap in Permanent vs Determinant Decompositions

- **Verdict:** INCONCLUSIVE
- **Bridge:** Schur-Weyl Duality $\times$ Boolean Circuit Size for Permanent Polynomial
- **Recorded:** 2026-05-12 01:34 UTC
- **Entry ID:** 40591186a3ea

Statement

For all $n \geq 2$, the plethysm coefficient $\lambda(\text{perm}_n)$ of the symmetric power $S^k(\text{perm}_n)$ exceeds $\lambda(\text{det}_m) + \varepsilon \cdot n^2$ for $m < n^{\{1.5\}}$, where $\varepsilon > 0$ is universal. This gap persists under linear substitutions preserving the polynomial's symmetry type.

Rationale

Plethysm coefficients encode symmetry constraints of polynomial representations. By linking them to circuit complexity, we may expose algebraic barriers to expressing permanent as a determinant-like circuit, leveraging representation-theoretic invariants resistant to natural proofs.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.19s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d9d81969.py", line 81, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d9d81969.py", line 58, in run_trial
    if lambda_perm_n > lambda_det_m + epsilon * n ** 2:
       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

OverflowError: int too large to convert to float

Judge reasoning

Test crashed with overflow error before producing data; cannot verify conjecture's validity | next: Implement arbitrary-precision arithmetic or numerical stability checks for large n

Symmetric Group Fourier Coefficients Inverse Proportional to ACC^0 Circuit Size for Sipser Function

- **Verdict:** INCONCLUSIVE
- **Bridge:** Representation Theory of Finite Groups $\times$ ACC^0 Circuit Size for Sipser Function
- **Recorded:** 2026-05-12 01:48 UTC
- **Entry ID:** 567baddf9ba5

Statement

For the Sipser function S_n on n variables, the maximum absolute Fourier coefficient over the symmetric group S_n 's irreducible representations is $\Theta(1/k)$, where k is the minimal ACC⁰ circuit size computing S_n . This holds for all $n \geq 2$.

Rationale

Symmetric group representations capture combinatorial symmetries of permutation-based functions like Sipser's. Fourier coefficients over these representations quantify 'non-abelian harmonic analysis' of the function, potentially revealing structural obstructions to efficient ACC⁰ computation.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_136fa469.py", line 80, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_136fa469.py", line 59, in run_trial
    for k in range(1, 2**n + 1):
                  ^
```

NameError: name 'n' is not defined

Judge reasoning

Test crashed due to undefined variable 'n' before producing results | next: Fix the NameError in test code by defining 'n' before usage

Integer Point Count in Solution Polytope Bounds Extended Frege Proof Length

- **Verdict:** INCONCLUSIVE
- **Bridge:** Geometry of Numbers × Extended Frege Proof Length
- **Recorded:** 2026-05-12 01:54 UTC
- **Entry ID:** 4798bdb0751b

Statement

For a 3-SAT instance Φ with n variables, the number of integer solutions $S(\Phi)$ satisfies $S(\Phi) \geq 2^{\{n - L(\Phi)\}}$, where $L(\Phi)$ is the length of the Extended Frege proof of Φ . Equality holds when Φ is a tautology with a unique solution.

Rationale

The geometry of numbers connects the combinatorial structure of integer solutions to algebraic constraints, while Extended Frege proofs formalize logical deductions. The conjecture posits that solution density directly impacts proof complexity via exponential trade-offs, exposing hidden geometric regularities in tautology refutations.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8072c074.py", line 82, in <module>
    result = run_trial(seed)
    ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8072c074.py", line 64, in run_trial
    S_phi = sum(1 for _ in filter(dpll, [sat_formula]))
    ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8072c074.py", line 64, in <genexpr>
    S_phi = sum(1 for _ in filter(dpll, [sat_formula]))
    ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8072c074.py", line 27, in dpll
    literal, rest = sat_formula[0]
    ~~~~~
```

ValueError: too many values to unpack (expected 2)

Judge reasoning

Test crashed during execution, preventing data collection. The error suggests a structural issue with the `sat_formula` handling in the DPLL algorithm. | next: Debug the DPLL implementation's handling of `sat_formula` structure to resolve the unpacking error

Matroid Rank Bounded by Nisan-Wigderson Seed Length for 3-SAT

- **Verdict:** INCONCLUSIVE
- **Bridge:** Matroid Theory $\times$ Nisan-Wigderson PRG Seed Length
- **Recorded:** 2026-05-12 02:14 UTC
- **Entry ID:** fa70962e79f5

Statement

For any 3-SAT instance with n variables, the seed length of the Nisan-Wigderson PRG fooling it is at most the rank of the matroid formed by its clause hypergraph, where the rank is the maximal size of a set of clauses with pairwise disjoint variable support.

Rationale

Matroid rank captures combinatorial dependencies between clauses, which may constrain the pseudorandomness needed to fool the formula. This bridges structural properties of CNF formulas with PRG efficiency, leveraging matroid theory's role in encoding combinatorial constraints.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 124, elapsed: 240.01s

TIMEOUT after 240s

Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Run test with extended timeout and memory limits to complete execution

Matroid Rank Submodularity and Monotone Circuit Size for k-CLIQUE

- **Verdict:** INCONCLUSIVE
- **Bridge:** Matroid Theory × Monotone Circuit Size for k-CLIQUE
- **Recorded:** 2026-05-12 02:20 UTC
- **Entry ID:** b052cc185466

Statement

For a k-CLIQUE CNF formula Φ on n variables, the rank function $r(S)$ of the matroid M constructed from Φ satisfies $r(S) \geq \Omega(n^{\{1/2\}})$ for any subset S of size $\Omega(n^{\{1/2\}})$, and the minimum number of elements needed to achieve $r(S) = \Omega(n^{\{1/2\}})$ is $\Omega(n^{\{1/2\}})$.

Rationale

The submodularity of the rank function in matroid theory could mirror the structure of monotone circuits, where dependencies between clauses (elements) affect the circuit's size. By linking the rank's growth rate to the circuit's complexity, we might uncover structural constraints.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: -1, elapsed: 0.00s

(no output)

Judge reasoning

Test crashed before producing data, preventing evaluation of support fraction or counterexamples. | next: Re-run test with detailed logging to identify crash cause and collect empirical evidence

Symplectic Rank of Disjointness Communication Matrix Bounds Randomized Complexity

- **Verdict:** INCONCLUSIVE
- **Bridge:** Symplectic Geometry × Communication Complexity of DISJOINTNESS
- **Recorded:** 2026-05-12 02:46 UTC
- **Entry ID:** 8acf355ada09

Statement

The symplectic rank of the DISJOINTNESS communication matrix M is $\Omega(n)$, where the symplectic rank is defined as the minimal k such that M decomposes into k rank-1 symplectic tensors under a fixed symplectic form. This lower bounds the randomized communication complexity of DISJOINTNESS.

Rationale

Symplectic geometry provides invariants that capture structural properties of matrices, which may correlate with communication complexity. The symplectic rank, as a decomposition invariant, could expose hidden algebraic constraints in the matrix that mirror information-theoretic barriers.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dfb495a0.py", line 102, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dfb495a0.py", line 74, in run_trial
    if not is_symplectic_form(M, n):
           ^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dfb495a0.py", line 44, in is_symplectic
    return matrix_multiply(matrix_multiply(M, M), I) == -I and matrix_multiply(transpose(M), M) == -
I
```

^^

TypeError: bad operand type for unary -: 'list'

Judge reasoning

Test crashed due to type error in matrix operations, preventing evaluation of conjecture | next: Fix matrix arithmetic implementation to handle symplectic form verification correctly

Schur Coefficient Density Inverse Proportional to Frege Proof Length

- **Verdict:** INCONCLUSIVE
- **Bridge:** Symmetric Function Theory $\times$ Frege Proof Length
- **Recorded:** 2026-05-12 03:10 UTC
- **Entry ID:** 2b3f03ad1b52

Statement

For a homogeneous polynomial f of degree d in n variables, let $S(f)$ denote the number of non-zero coefficients in its Schur decomposition. Then, the Frege proof length of $f = 0$ is at least $\Omega(S(f)^{1/2})$

Rationale

The density of Schur coefficients reflects the complexity of the polynomial's representation, which might correlate with the difficulty of proving its identity through Frege systems. Symmetric function theory provides a framework to decompose polynomials into irreducible components, exposing structural properties that could influence proof complexity.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.10s

```
ue': 1, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': ''}
TRIAL: {'metric_name': 'Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
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TRIAL: {'metric_name': 'Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
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TRIAL: {'metric_name': 'Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
```

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_98998d42.py", line 121, in <module>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                        ~~~~~^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_98998d42.py", line 121, in <genexpr>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                        ~~~~~^
```

KeyError: 'seed'

Judge reasoning

Test crashed due to KeyError before producing reliable data | next: Fix the test code to handle missing 'seed' field and re-run with proper error handling

Free Cumulant Sum Bounded by Log-Size for Read-Twice BPs

- **Verdict:** INCONCLUSIVE
- **Bridge:** Free Probability Theory × Read-Twice Branching Programs
- **Recorded:** 2026-05-12 03:25 UTC
- **Entry ID:** 93b47d86db79

Statement

For any read-twice BP P computing IP_2 , the sum of free cumulants $\sum_{k=1}^{\infty} \kappa_k(P) \leq \log(\text{size}(P)) + O(1)$. For random read-twice BPs over $GF(2)$, $\sum \kappa_k(P) = \Theta(\log n)$.

Rationale

Free cumulants capture non-commutative independence structures; their sum could distinguish read-twice BPs (with limited independence) from unrestricted BPs (with full independence). IP_2's inherent symmetry might force linear cumulant growth.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 124, elapsed: 240.01s

TIMEOUT after 240s

Judge reasoning

Test timed out without producing supporting data or counterexamples | next: Increase timeout duration and re-run tests with extended computation limits

Condition Number of Moment Matrix Inversely Proportional to SOS Degree for Max-CUT

- **Verdict:** INCONCLUSIVE
- **Bridge:** Convex Geometry (Condition Number of Moment Matrices) $\times$ SOS Degree for Max-CUT
- **Recorded:** 2026-05-12 04:34 UTC
- **Entry ID:** f89319bf92ad

Statement

For a random max-CUT instance with n variables, the condition number κ of the moment matrix in the degree- d SOS relaxation satisfies $\kappa = \Theta(1/d^2)$, and the SOS degree required to achieve an approximation ratio of $0.878 - \varepsilon$ is $\Omega(\sqrt{d})$

Rationale

The condition number of the moment matrix reflects the sensitivity of the SOS relaxation to perturbations. A lower condition number implies better-conditioned matrices, which may require lower SOS degrees to achieve the same approximation ratio. This link could reveal geometric properties of the feasible region in the SOS hierarchy

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.10s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cb36e3f3.py", line 92, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```



```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cb36e3f3.py", line 65, in run_trial
    inv_M = matrix_inv(M)
            ^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cb36e3f3.py", line 42, in matrix_inv
    A_rref = gaussian_elimination(A_augmented)
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cb36e3f3.py", line 28, in gaussian_elim
    raise ValueError("Matrix is singular")
ValueError: Matrix is singular

```

Judge reasoning

Test crashed with singular matrix error, preventing validation of conjecture | next: Investigate why matrix is singular for tested parameters and verify matrix construction

Projective Plane Incidence Bounds Nisan-Wigderson Seed Length

- **Verdict:** INCONCLUSIVE
- **Bridge:** Finite Geometry (Projective Planes) × Nisan-Wigderson PRG Seed Length
- **Recorded:** 2026-05-12 04:40 UTC
- **Entry ID:** c9054923e964

Statement

For a CNF Φ with n variables whose clause incidence matrix is a projective plane of order q , the Nisan-Wigderson seed length for Φ is $\leq q^2 + q + 1$. Conversely, if Φ has seed length $> q^2 + q + 1$, its incidence matrix cannot be a projective plane of order q .

Rationale

Projective planes exhibit optimal combinatorial expansion properties that could constrain pseudo-random generator seed requirements. The bound $q^2 + q + 1$ matches the number of lines in the plane, suggesting a direct geometric constraint on the PRG's pseudorandomness threshold.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

```

• exit code: 1, elapsed: 0.09s
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_54288a4f.py", line 148, in <module>
    result = run_trial(seed)
              ^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_54288a4f.py", line 110, in run_trial
    cnf = generate_cnf_from_plane(points, lines)
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_54288a4f.py", line 85, in generate_cnf_
    clause = [j + 1 for j in range(m) if (i // q, i % q) in lines[j]]
              ^
NameError: name 'q' is not defined

```

Judge reasoning

Test crashed due to undefined variable 'q' in the generation logic, preventing validation of the conjecture | next: Fix the code to properly define 'q' in the projective plane generation and re-run tests

Fourier Coefficient Spread and ACC⁰ Circuit Lower Bounds

- **Verdict:** INCONCLUSIVE
- **Bridge:** Fourier Analysis on Finite Groups × ACC⁰ Circuit Size
- **Recorded:** 2026-05-12 05:17 UTC
- **Entry ID:** 90abe309451a

Statement

For any boolean function $f: \{0,1\}^n \rightarrow \{0,1\}$, if the sum of absolute Fourier coefficients $\sum |\hat{f}(S)| \geq 2^{\{n/2\}} \sqrt{n}$, then f cannot be computed by an ACC⁰ circuit of size $2^{\{n/2\}}$.

Rationale

The spread of Fourier coefficients reflects correlation with parity functions. ACC⁰ circuits, limited to bounded-depth polynomial-size gates, struggle to capture such correlations, making large Fourier sums a barrier to efficient computation.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 0.07s

```
tric_name': 'Fourier Coefficient Sum', 'metric_value': 0.64453125, 'instances_tested': 1, 'conjecture': 1
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.626953125, 'instances_tested': 1, 'conjecture': 1}
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.59765625, 'instances_tested': 1, 'conjecture': 1}
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.5625, 'instances_tested': 1, 'conjecture': 1}
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.634765625, 'instances_tested': 1, 'conjecture': 1}
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.625, 'instances_tested': 1, 'conjecture': 1}
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.546875, 'instances_tested': 1, 'conjecture': 1}
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.56640625, 'instances_tested': 1, 'conjecture': 1}
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.5546875, 'instances_tested': 1, 'conjecture': 1}
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.59765625, 'instances_tested': 1, 'conjecture': 1}
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.658203125, 'instances_tested': 1, 'conjecture': 1}
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.625, 'instances_tested': 1, 'conjecture': 1}
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.666015625, 'instances_tested': 1, 'conjecture': 1}
RESULT: SUPPORTED mean=0.6087239583333334 std=0.03415580763030604 support_fraction=1.0
```

Judge reasoning

The test used $n=1$ where the conjecture's threshold ($2^{\{n/2\}} \sqrt{n}$) is ~ 1.414 , but all metric values (< 1) never reached this threshold. Thus, the conjecture's condition was never tested. | next: Test with $n \geq 2$ where $2^{\{n/2\}} \sqrt{n} \geq 2$ to validate the conjecture's actual condition

Fourier Coefficient Decay Bounds AC^0 Circuit Size for Parity-Insensitive Functions

- **Verdict:** INCONCLUSIVE
- **Bridge:** Fourier Analysis on Finite Groups $\times$ AC^0 Circuit Size
- **Recorded:** 2026-05-12 05:43 UTC
- **Entry ID:** 29d6f7d21e9f

Statement

For any parity-insensitive Boolean function $f: \{0,1\}^n \rightarrow \{0,1\}$, the sum of absolute Fourier coefficients $\sum_{S \neq \emptyset} |\hat{f}(S)|$ decays exponentially with n iff the minimal AC^0 circuit size computing f is $\Omega(2^{n/2})$.

Rationale

Parity-insensitive functions exhibit Fourier coefficient concentration patterns that may correlate with algebraic complexity. The exponential decay threshold could expose a structural gap between Fourier analytic simplicity and circuit complexity, leveraging the inherent symmetry of AC^0 computation.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.29s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0b03edf7.py", line 97, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0b03edf7.py", line 67, in run_trial
    f_hat = compute_fourier_coefficients(clauses, n)
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0b03edf7.py", line 38, in compute_fourier_coefficients
    return [abs(coeff / (1 << n)) for coeff in fast_walsh_hadamard_transform(f)]
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0b03edf7.py", line 23, in fast_walsh_hadamard_transform
    f[j] += f[j + n // 2]
           ~^^^^^^^^^^^^
```

IndexError: list index out of range

Judge reasoning

Test crashed with IndexError before producing results, preventing evaluation of conjecture validity
| next: Debug fast_walsh_hadamard_transform implementation to fix index out of range error

Invariant Ring Degree Bounds SOS Refutation Degree for Symmetric CSPs

- **Verdict:** INCONCLUSIVE
- **Bridge:** Invariant Theory $\times$ SOS Refutation Degree for CSPs
- **Recorded:** 2026-05-12 06:36 UTC

- **Entry ID:** 3a288632d238

Statement

For a symmetric CSP instance Φ invariant under a finite group G , the minimal degree d of a generating set of the invariant ring of G is inversely proportional to the SOS refutation degree of Φ . Specifically, the SOS refutation degree is at least $\Omega(1/d)$.

Rationale

Symmetric CSPs can be analyzed via their invariant rings, which capture the group's action. The degree of the generators may limit the SOS hierarchy's ability to refute the instance, as higher-degree generators might require higher SOS levels.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9d483de9.py", line 74, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9d483de9.py", line 51, in run_trial
    instance = generate_symmetric_csp(3) # Example: n=3
              ^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9d483de9.py", line 37, in generate_symmetric_csp
    while not is_symmetric(instance):
            ^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9d483de9.py", line 30, in is_symmetric
    permuted_instance = [instance[g[i]] for i in range(len(instance))]
                        ~^^^
```

IndexError: list index out of range

Judge reasoning

Test crashed with IndexError before producing results, preventing evaluation of conjecture validity.
| next: Fix the index out-of-range error in generate_symmetric_csp() and re-run tests with multiple seeds

Noncommutative L^p Norm of Communication Matrix Bounds Randomized Disjointness Complexity

- **Verdict:** INCONCLUSIVE
- **Bridge:** Noncommutative L^p Geometry $\times$ Communication Complexity of DISJOINTNESS
- **Recorded:** 2026-05-12 06:43 UTC
- **Entry ID:** cfd6712a7d9a

Statement

For a random instance M of DISJOINTNESS on n elements, the noncommutative L^p norm $\|M\|_p \geq \Omega(\sqrt{n})$ for $p=2$, with equality achieved when M is a balanced bipartite graph with uniform edge weights.

Rationale

Noncommutative L^p norms capture operator-theoretic structure of tensors, which may encode entanglement-like constraints in communication protocols. The L^2 norm's relation to matrix rank could expose hidden geometric barriers in randomized protocols for DISJOINTNESS.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 0.10s

```
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RESULT: SUPPORTED mean=28.284271247461902 std=undefined support_fraction=1.0
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Judge reasoning

Safety rail: critic_challenge | original judge: | next: NONE